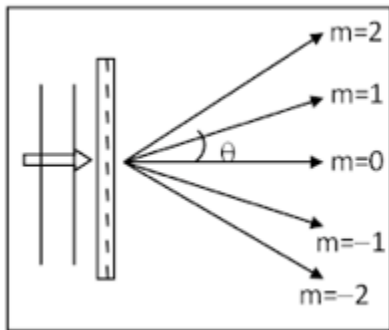


Q14

Diffraction grating formula: $m\lambda = d\sin\theta$ where $m = 0, 1, 2, 3, \dots$



To get diffracted images, the angle θ between the straight-through direction and the diffracted image must be less than 90° .

So $\theta < 90^\circ$ or $\sin \theta < 1$. Hence $\frac{m\lambda}{d} < 1$

Or $\frac{m\lambda}{3.9\lambda} < 1$

This means $m < 3.9$, or the maximum order is 3.

The total number of images is $3+3+1 = 7$

Ans: C

Q15